

SAHANA ARUMUGAM

Chennai, Tamil Nadu, India | Open to Remote & Onsite | +91 90424 12219
sahanaarums16@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Penultimate-year Integrated M.Tech student in Software Engineering at VIT Vellore (CGPA: 9.14) with strong foundations in data structures, algorithms, object-oriented programming, and distributed systems. Experienced in building scalable software solutions, concurrent applications, and high-performance systems through industry and research projects. Proficient in Python, Java, and C++, with hands-on experience in backend development and cloud technologies. Solved 300+ DSA problems on LeetCode and passionate about building reliable, efficient software at scale.

Education

Vellore Institute of Technology (VIT), Vellore

2023–2028 (Expected)

Integrated M.Tech in Software Engineering

CGPA: 9.14

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Compiler Design, Distributed Systems, Database Systems, Parallel Computing, Computer Networks, Machine Learning

Technical Skills

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, R

Systems Programming & Distributed Computing: Distributed Computing, Concurrent Programming, Multi-threading, Synchronization, System Design, Edge Computing (Raspberry Pi), GCC, LLVM

Data Structures, Algorithms & Software Design: Data Structures, Algorithms, Object-Oriented Programming (OOP), Software Design, Design Patterns, UML Diagrams, Performance Optimization, Debugging

Software Testing & Quality Assurance: JUnit, Selenium, Mocha, Unit Testing, Integration Testing, Functional Testing, System Testing, Regression Testing, Test Case Design

Machine Learning, AI & Data Science: TensorFlow, PyTorch, Scikit-learn, XGBoost, Bayesian Optimization, NumPy, Pandas, Predictive Modeling, Data Mining, Exploratory Data Analysis (EDA), Matplotlib, Seaborn

Backend Development & Databases: Node.js, Express.js, REST APIs, MongoDB, SQL, Firebase, Supabase, Oracle

Cloud, DevOps & Deployment: AWS, Azure, Docker, Jenkins, GitHub Actions, Vercel, Render

Experience

AI/ML Intern — Team Lambda (DST-Funded Research)

Dec 2025 – June 2026

- Engineered concurrent ML/DL pipelines using TensorFlow, PyTorch, and Scikit-learn across 20+ weather and operational datasets (100K+ rows), reducing model iteration time by approximately 40%.
- Designed a distributed edge-processing system using Raspberry Pi nodes for real-time monitoring of environmental conditions and solar panel telemetry, achieving less than 50ms processing latency.
- Developed 3+ predictive models leveraging weather forecasts, dust accumulation patterns, and power-generation metrics, achieving 88–92% prediction accuracy for cleaning schedule optimization.
- Delivered end-to-end ML solutions for DST-funded research, improving cleaning efficiency by approximately 25%, reducing unnecessary cleaning cycles by 30%, and supporting large-scale performance evaluation.

IEEE Information Theory Society — Senior Technical Team Member

Jan 2025 – Nov 2025

- Implemented 8+ features using React and Node.js applying software design and reliability engineering principles, reducing usability issues by approximately 60% across 500+ users.
- Mentored 50+ participants in data structures, algorithms, and system design across ML workshops and coding contests.

Projects

MoVITex — AI-Powered Campus Crowd Intelligence Platform

[GitHub](#)

React, TypeScript, Python, Scikit-learn

- Built a crowd intelligence platform with concurrent data pipelines processing 50+ zones and 10K+ daily events.
- Engineered multi-threaded congestion classification achieving approximately 91% accuracy, reducing peak-hour errors by 35%.
- Built real-time dashboards with smart alerting delivering less than 2-second latency through performance optimization.

Bayesian Compiler Optimization Engine

[GitHub](#)

Python, Scikit-learn, XGBoost, GCC, LLVM, Bayesian Optimization

- Designed a machine learning driven compiler optimization framework predicting optimal GCC/LLVM transformation strategies across 20+ configurations.
- Engineered concurrent experimentation pipelines with synchronized multi-threaded workers, achieving approximately 15% runtime improvement and 40% faster optimization search.

GlassForge Sentinel — Predictive Maintenance Platform

[GitHub](#)

Python, RAG, Blockchain, Next.js

- Architected a distributed predictive maintenance platform with fault-tolerant pipelines processing multi-sensor industrial telemetry in real time.
- Built ML failure prediction models achieving 88–92% accuracy, reducing downtime by approximately 30%.
- Integrated RAG-powered LLMs and blockchain smart contracts across distributed systems.

Achievements

1st Place — HackHerThon, SVCE Chennai (National-Level Hackathon)

Designed and built an ML-driven predictive maintenance system integrating RAG-based LLMs and blockchain for explainable, tamper-proof diagnostics; selected best among 100+ teams nationwide.

1st Place — IEEE TEMS Hack Expertise Hackathon, VIT Vellore

Architected and delivered a full-stack sustainability-focused exchange platform under rapid prototyping constraints.

Track Winner — IEEE ITS Hackathon, VIT Vellore (Campus Innovation Track)

Built a real-time crowd management system with ML-driven predictive routing for a campus population exceeding 20,000.

Proficiency in Artificial Intelligence — Academic Recognition

Awarded for academic excellence in Artificial Intelligence and Machine Learning specialization coursework.